YOLO Based Recognition of Distracted Driving from In-Cabin Images

Caleb Bloch

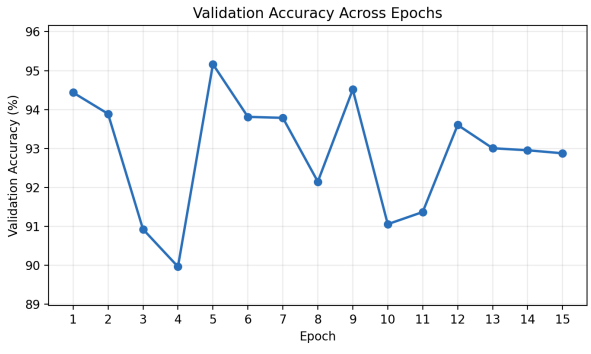
Project Overview. This project studies distracted driving recognition from in-cabin images using a YOLO image classification model. For Level 1, I simplified the task to binary classification, where the model predicts whether a driver is focused or distracted. I used the State Farm Distracted Driver Detection dataset and created a fixed train/validation split so the same held-out data could be reused across experiments. I then used some built in YOLO augmentationss to increase the robustness of the model. The goal of this stage was to build a complete end-to-end pipeline from preprocessing and training to evaluation and basic qualitative analysis.

Level 1 Goals. My Level 1 goals were to:

- Create a persistent binary train/validation split.
- Treat c0 as focused and group c1 through c9 into distracted.
- Train a YOLO classification model on the binary task.
- Calculate validation accuracy, precision, recall, and F1 score.
- Inspect a few labeled validation cases for basic error analysis.

Results So Far. The current binary model achieved 95.14% accuracy on the validation split, with a weighted F1 score of 0.9483. Because the binary dataset is heavily imbalanced toward distracted-driving images, accuracy alone doesn't tell the whole story: a naive model that predicts distracted for every validation image would already reach about 88.1% accuracy. For that reason, precision, recall, and F1 score are more important metrics here. Performance was stronger on the distracted class than on the focused class: distracted examples had precision 0.9569, recall 0.9894, and F1 0.9729, while focused examples had precision 0.8947, recall 0.6696, and F1 0.7660.

- Validation accuracy: 95.14% (majority-class baseline: 88.12%)
- Focused: precision 0.8947, recall 0.6696, F1 0.7660
- Distracted: precision 0.9569, recall 0.9894, F1 0.9729
- Weighted F1: 0.9483
- Confusion matrix counts: 306 focused correct, 151 focused predicted as distracted, 36 distracted predicted as focused, 3353 distracted correct



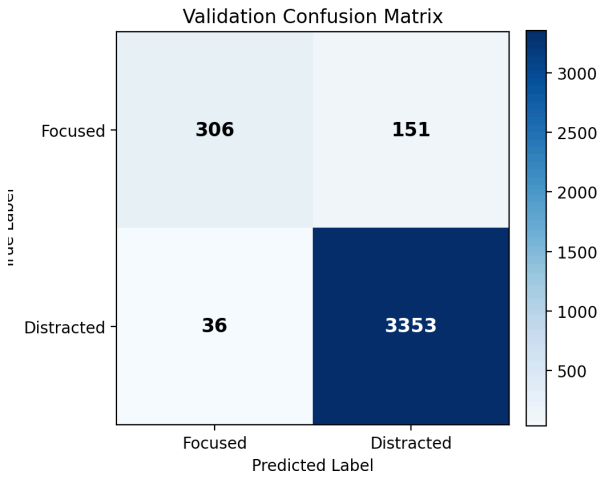


Figure 1. Validation accuracy across epochs. The best validation accuracy appears at epoch 5, and later epochs do not clearly improve on that point.

Figure 2. Confusion matrix on the held-out validation split. The model identifies distracted driving very well, but it misses a noticeable number of focused examples.

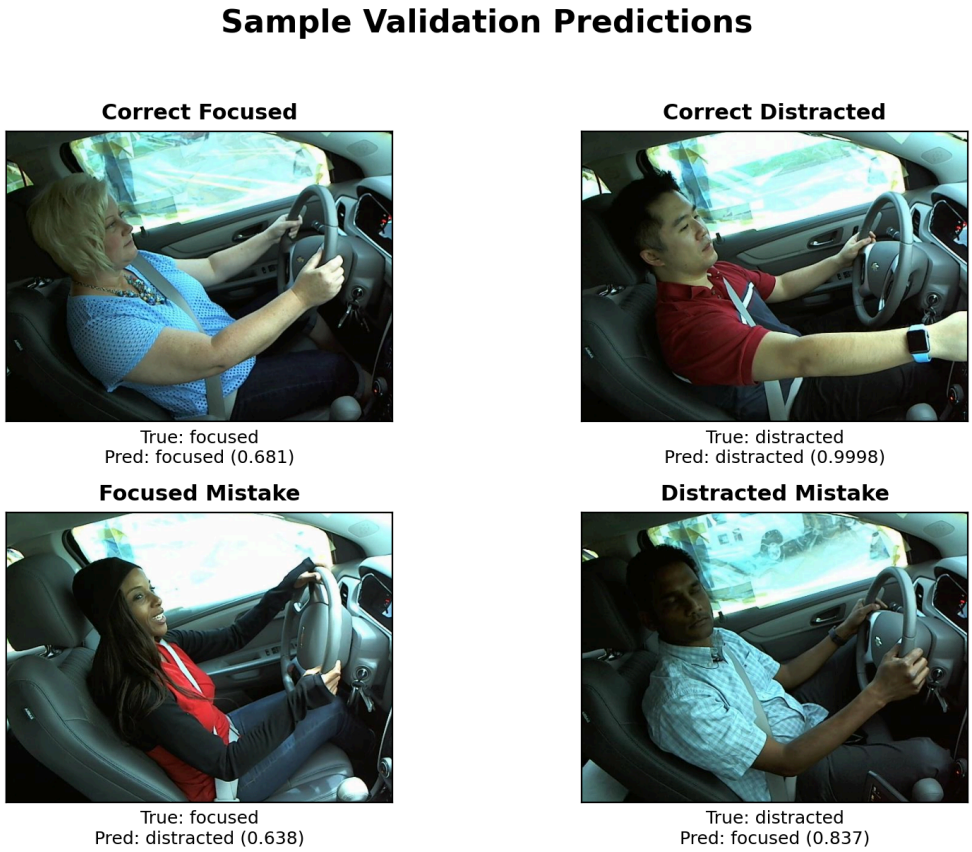


Figure 3. Sample labeled validation cases. These examples show two correct predictions and two mistakes from the held-out validation set. The mistaken images in particular are good examples of edge cases that even a human labeler would likely feel uncertain about their classification.

Interpretation. These results suggest that the binary version of the problem is workable with a small YOLO classifier, especially for identifying distracted drivers. The focused-driving recall is still noticeably weaker, which makes sense given the class imbalance and the fact that some driver poses are visually similar. One concern I have is the model's bias towards distracted classifications due to the bias in its training data. If I were to do this again, I'd use training data more evenly split between the two binary classifications. Overall, the Level 1 pipeline works end to end and produces clear, interpretable results on held-out data.